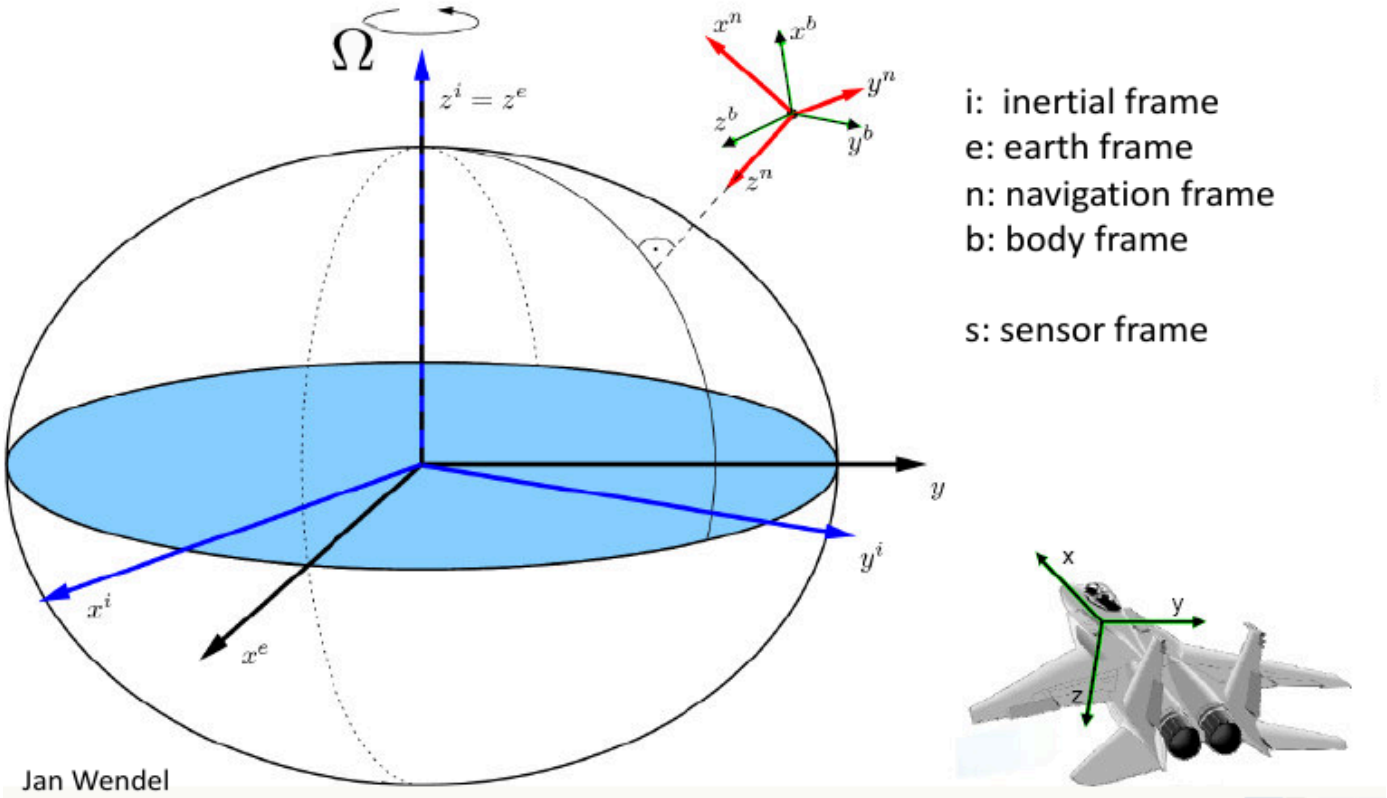
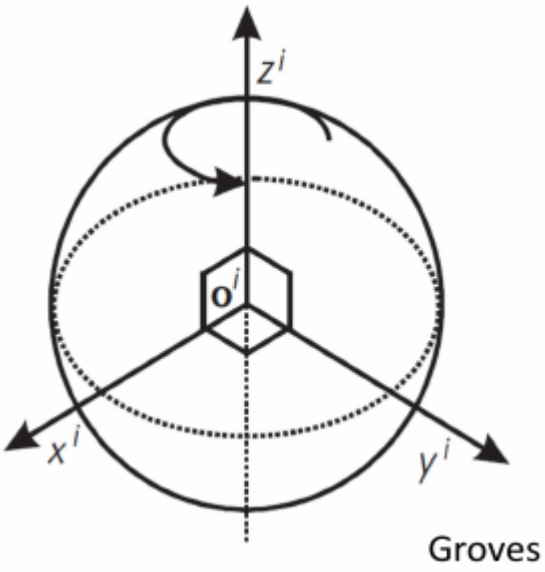
Ms03: Relative Sensors

Coordinate Frames

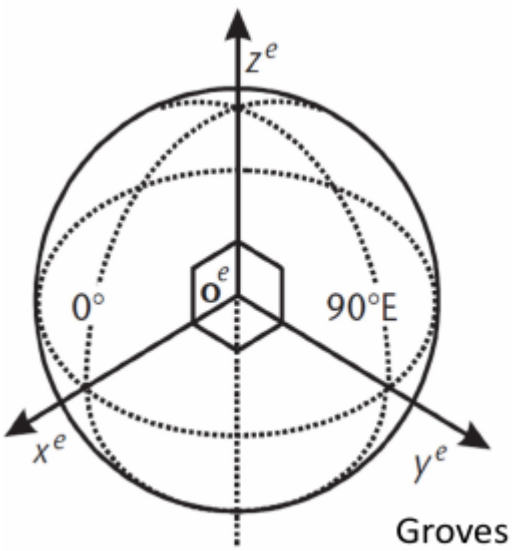


Inertial Frame (I-Frame)

- **Inertial Frame**:
  - Frame without any acceleration or rotation with respect to the rest of the universe
- **Earth-Centered Inertial Frame (ECIF)**
  - Origin: Centre of mass of the Earth
  - z-axis: along the rotational axis through North
  - x,y-axes: in equatorial plane, fixed in space
- **Inertial sensors measure with respect to the inertial frame**



Earth Centred Earth Fixed Frame (ECEF) (E-Frame)

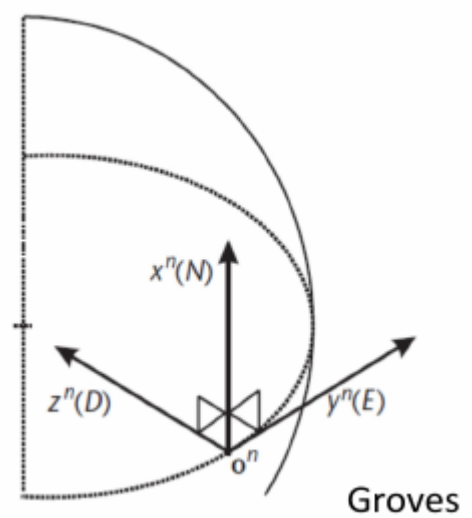


-
- Similar to i-frame, but with x,y-axes also fixed to Earth
  - Origin: Center of mass = Earth center of mass
  - z-axis: along rotational axis ($z^e = z^i$) through North
  - x-axis: intersection of equatorial plane and reference meridian (Greenwich, zero latitude)
  - y-axis: completes right-handed system
- **E-frame rotates to the i-frame** along its **z-axis** with the **Earth Rotation Rate**
- **User position and velocity information** are usually given with respect to the **e-frame**

Local Navigation Frame (N-Frame)

- Local system tangential to the ellipsoid of Earth model
  - origin: reference point of the user
    - e.g. centre of vehicle mass
  - z-axis: pointing down parallel to gravity

    - x-axis: towards North
    - y-axis: Completes right-handed system (East)
- Coordinates given as NED (North East Down)
    - Alternative defintions
- **N-frame** is usually used as a reference system for **attitude information**

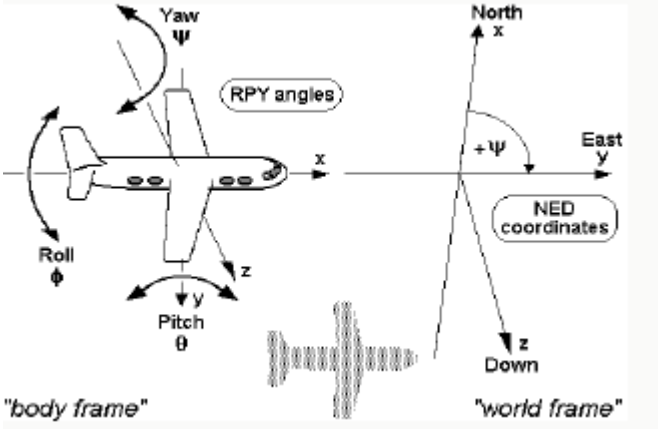


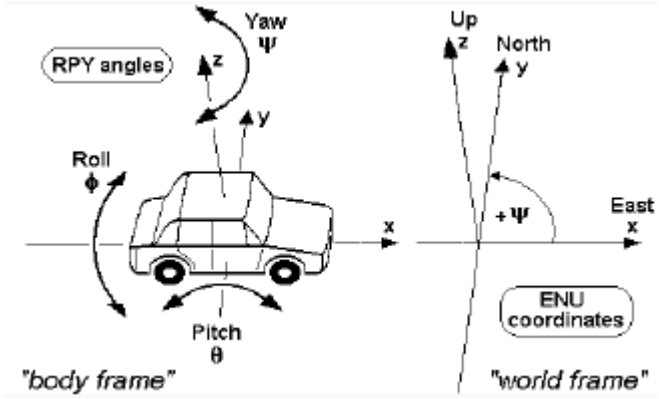
Body Frame /Vehicle Frame (B-Frame)

- Attached to the user (e.g. vehicle)
    - origin: point on vehicle
        - e.g. centre of mass of vehicle
    - z-axis: pointing down
    - x-axis: pointing forward
    - y-axis: pointing to the right, completing the RHS
- Alternative definitions
- **Attitude** (Orientation) of a system is usually given as the **orientation** of the **b-fame relative to the n-frame**
- **Sensor data** are assumed to be available in the **b-frame**
    - (System Calibration!)

Ground based/Aerial Vehicles

- Ground based and aerial vehicles usually have different definitions of the b- and n-frame
- **Aerial**:
    - n-frame: North-East-Down
    - b-frame:Forward-Right-Down
    - -> Heading towards: East: $\Psi = 90^\circ$

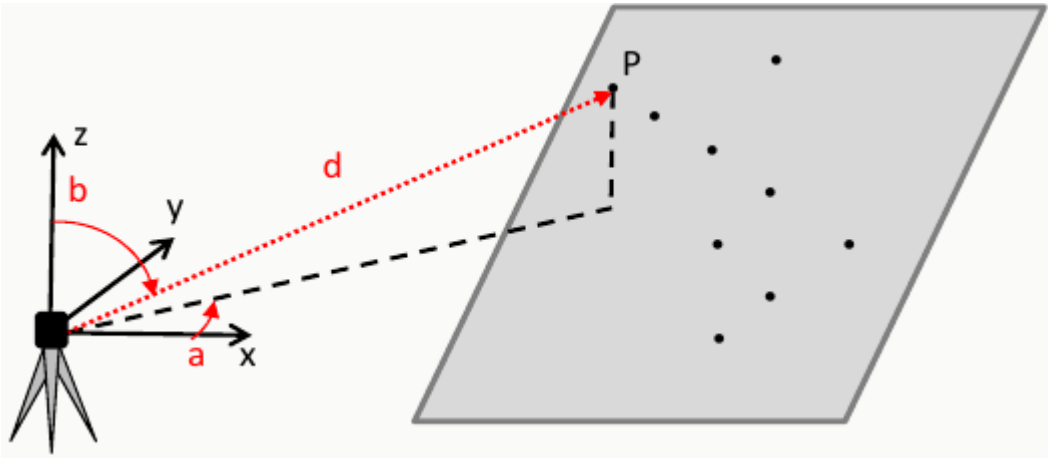
- 

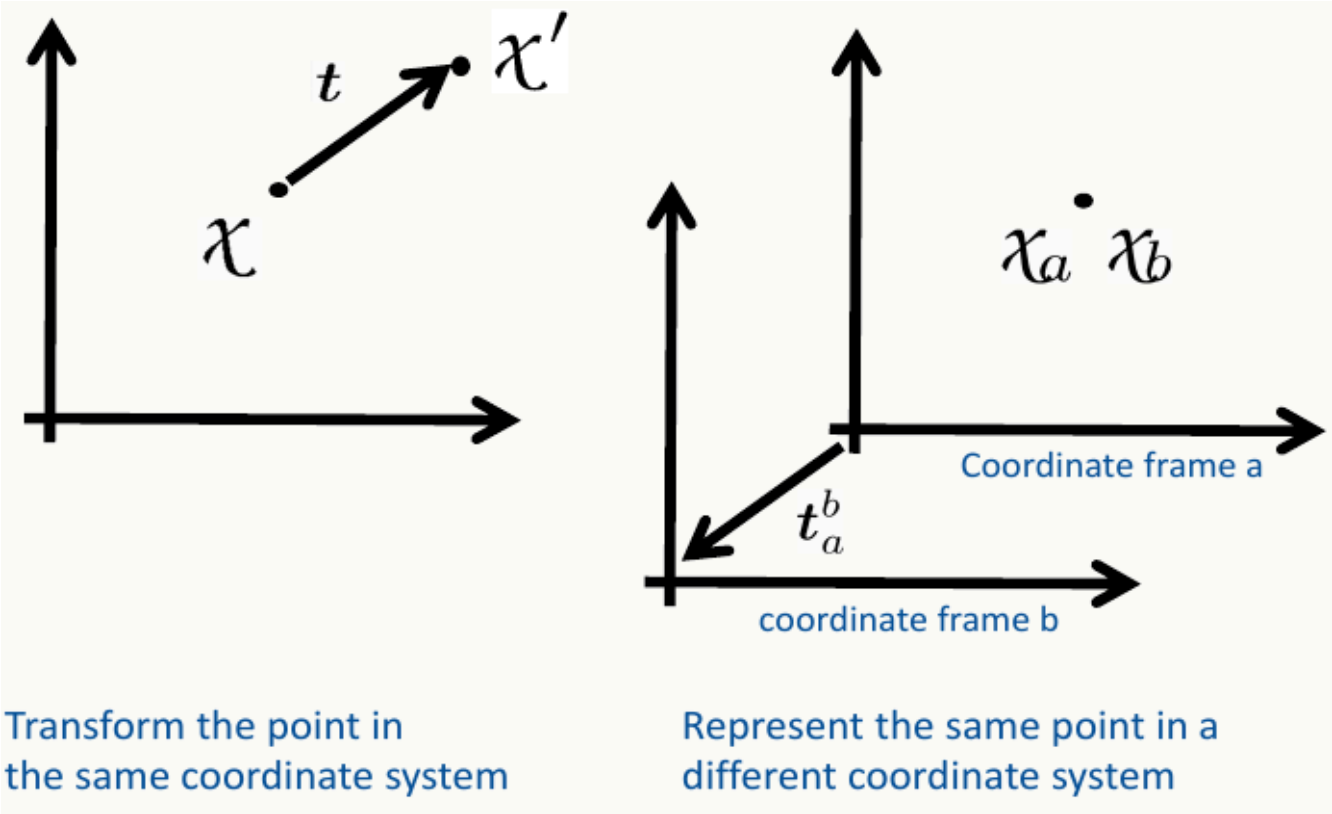
- **Ground based**:
  - n-frame: East-North-Up
  - b-frame: Forward-left-up
  - Heading towards North: $\Psi = 90^\circ$
  - 

- All **angles zero** means **no rotation between b- and n-frame**
- *RHS-rule for both, but*
  - *direction of b-framefor aerial towards earth*
  - *North, East components changed for n-frame*

Sensor Frame (S-Frame)

- S-Frame is the frame, in which sensor data is given
  - Origin and axis specific to the sensor
- Example: Panoramic Laser Scanner

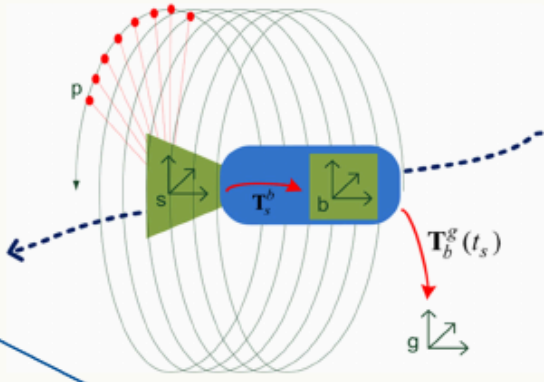
- 

- 

Georeferencing Measurement Data

- Chain of Transformations between coordinate frames

- $$\mathbf{p}_o^g = \mathbf{T}_b^g(t_s)\mathbf{T}_s^b\mathbf{p}_o^s(t_s)$$



$$\begin{bmatrix} \mathbf{R}_b^g & \mathbf{t}_b^g \\ \mathbf{0}^\mathsf{T} & 1 \end{bmatrix}\begin{bmatrix} \mathbf{R}_s^b & \mathbf{t}_s^b \\ \mathbf{0}^\mathsf{T} & 1 \end{bmatrix}\begin{bmatrix} \mathbf{p}_o^s \\ 1 \end{bmatrix}$$

Kinematic Quantities

- Unambiguous definition of kinematic quantities such as position, velocity, acceleration and angular rate
  - velocity of the **b-frame**, relative to the **e-frame**, given in coordinates of the **n-frame**
    - $\boldsymbol{v}_{eb}^n$
  - Change of reference:
    - $\boldsymbol{v}_{eb}^n = -\boldsymbol{v}_{be}^n$
    - $\boldsymbol{\omega}_{eb}^n = -\boldsymbol{\omega}_{be}^n$
  - Concatenation:
    - $\boldsymbol{\omega}_{ib}^b = \boldsymbol{\omega}_{ie}^b + \boldsymbol{\omega}_{en}^b + \boldsymbol{\omega}_{nb}^b$
  - Change of resolving frame:
    - $\boldsymbol{\omega}_{nb}^b = \boldsymbol{C}_n^b \boldsymbol{\omega}_{nb}^n$

Rotation Matrices

- Euler angles rotation rotate from n-frame to b-frame
  - 3-2-1 convention

$$\begin{aligned} \mathbf{C}_n^b &= \underbrace{\mathbf{R}_1(\phi)}_{\text{roll}} \cdot \underbrace{\mathbf{R}_2(\theta)}_{\text{pitch}} \cdot \underbrace{\mathbf{R}_3(\psi)}_{\text{yaw}} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & \sin\phi \\ 0 & -\sin\phi & \cos\phi \end{bmatrix} \cdot \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix} \cdot \begin{bmatrix} \cos\psi & \sin\psi & 0 \\ -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

- The **Direction-Cosine-Matrix (DCM)** describes the rotations from the b to the n-frame

- $$\begin{aligned}\mathbf{C}_b^n &= (\mathbf{C}_n^b)^T = \mathbf{R}_3(\psi)^T \cdot \mathbf{R}_2(\theta)^T \cdot \mathbf{R}_1(\phi)^T \\ &= \begin{bmatrix} \cos\theta\cos\psi & -\cos\phi\sin\psi + \sin\phi\sin\theta\cos\psi & \sin\phi\sin\psi + \cos\phi\sin\theta\cos\psi \\ \cos\theta\sin\psi & \cos\phi\cos\psi + \sin\phi\sin\theta\sin\psi & -\sin\phi\cos\psi + \cos\phi\sin\theta\sin\psi \\ -\sin\theta & \sin\phi\cos\theta & \cos\phi\cos\theta \end{bmatrix}\end{aligned}$$

- Alternative: Quaternions
  - Quaternion -> DCM
    - $$\mathbf{C}_b^n = \begin{bmatrix} (w^2 + v_1^2 - v_2^2 - v_3^2) & 2(v_1v_2 - wv_3) & 2(v_1v_3 + wv_2) \\ 2(v_1v_2 + wv_3) & (w^2 - v_1^2 + v_2^2 - v_3^2) & 2(v_2v_3 - wv_1) \\ 2(v_1v_3 - wv_2) & 2(v_2v_3 + wv_1) & (w^2 - v_1^2 - v_2^2 + v_3^2) \end{bmatrix}$$
  - DCM-> Euler Angles
    - $$\mathbf{C}_b^n = \begin{bmatrix} (w^2 + v_1^2 - v_2^2 - v_3^2) & 2(v_1v_2 - wv_3) & 2(v_1v_3 + wv_2) \\ 2(v_1v_2 + wv_3) & (w^2 - v_1^2 + v_2^2 - v_3^2) & 2(v_2v_3 - wv_1) \\ 2(v_1v_3 - wv_2) & 2(v_2v_3 + wv_1) & (w^2 - v_1^2 - v_2^2 + v_3^2) \end{bmatrix}$$

Quaternions:

Rotation vector
(direction: rot. axis, length: rot. angle)

$$\mathbf{q}_b^n = \begin{pmatrix} w \\ v_1 \\ v_2 \\ v_3 \end{pmatrix} = [w, \mathbf{v}] = \left[\cos\frac{||\boldsymbol{\sigma}||}{2}, \frac{\boldsymbol{\sigma}}{||\boldsymbol{\sigma}||}\sin\frac{||\boldsymbol{\sigma}||}{2}\right] \equiv R\left(\frac{\boldsymbol{\sigma}}{||\boldsymbol{\sigma}||}, ||\boldsymbol{\sigma}||\right)$$

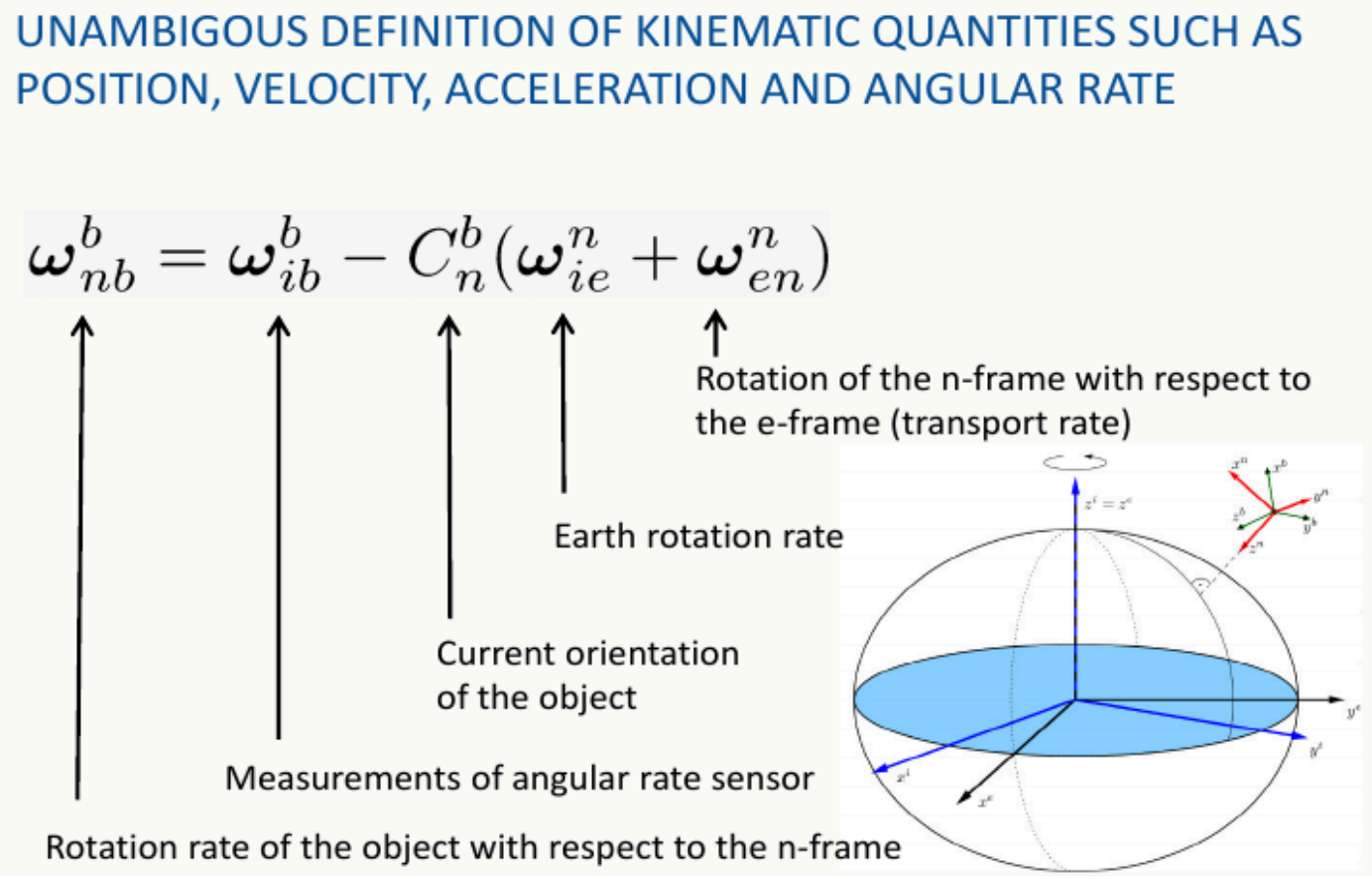
$$\begin{aligned} \text{(Addition)} \quad & \mathbf{q} + \mathbf{q}' = [w + w', \mathbf{v} + \mathbf{v}'] \\ \text{(Multiplication)} \quad & \mathbf{q} \cdot \mathbf{q}' = [ww' - \langle \mathbf{v}, \mathbf{v}' \rangle, w\mathbf{v}' + w'\mathbf{v} - \mathbf{v} \times \mathbf{v}'] \\ \text{(Konjugation)} \quad & \mathbf{q}^* = [w, -\mathbf{v}] \\ \text{(Norm)} \quad & ||q|| = \sqrt{\mathbf{q} \cdot \mathbf{q}^*} = \sqrt{w^2 + ||\mathbf{v}||^2} \\ \text{(Inverse)} \quad & \mathbf{q}^{-1} = \frac{\mathbf{q}^*}{||\mathbf{q}||^2}. \end{aligned}$$

=1 for Rotations

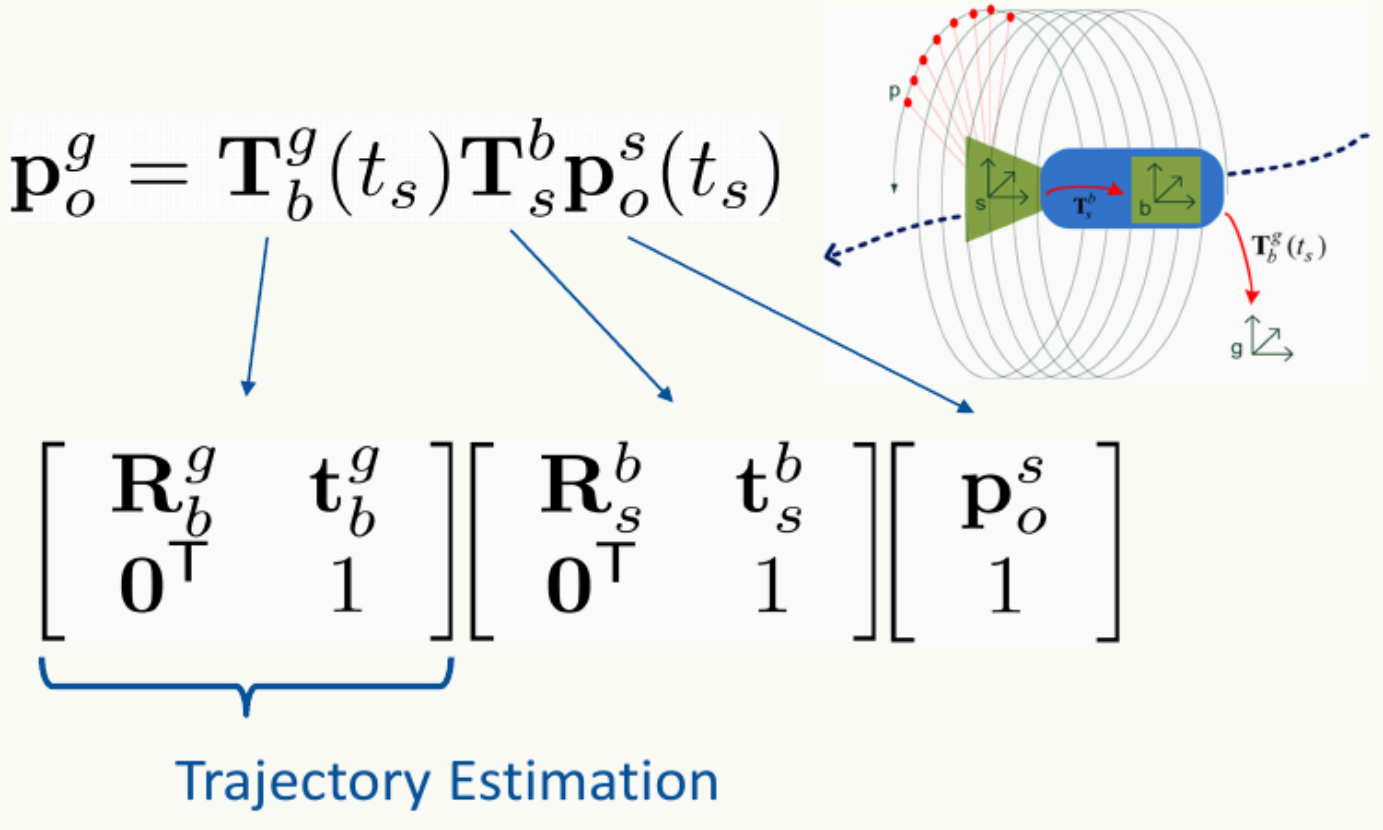
= Konjugation for Rotations

$$\begin{pmatrix} 0 \\ \mathbf{v}^n \end{pmatrix} = \mathbf{q}_b^n \cdot \begin{pmatrix} 0 \\ \mathbf{v}^b \end{pmatrix} \cdot \mathbf{q}_n^b$$

Rotation of a vektor

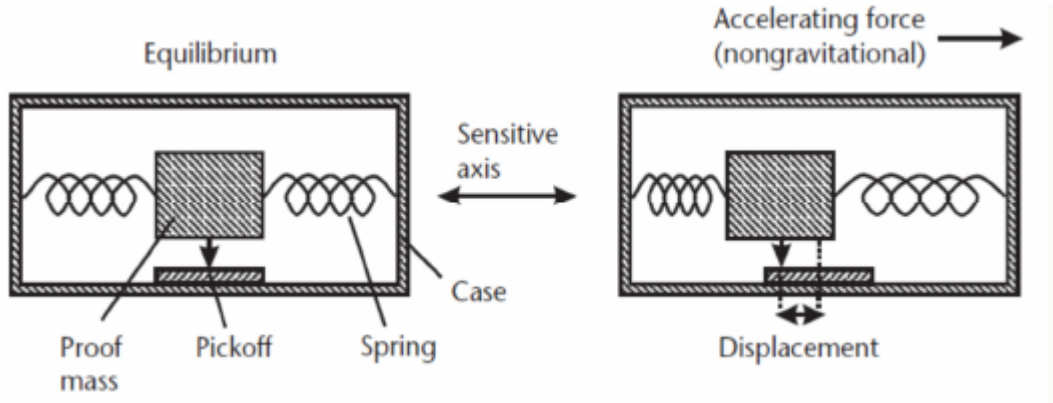


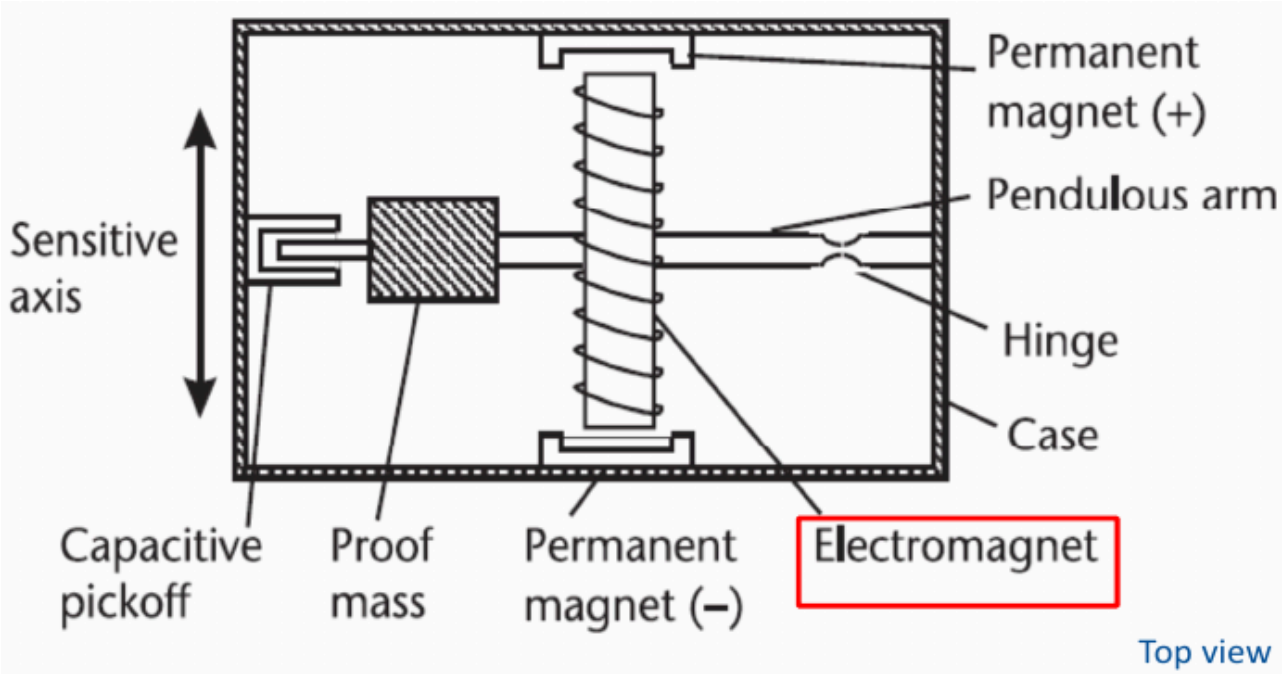
Trajectory Estimation



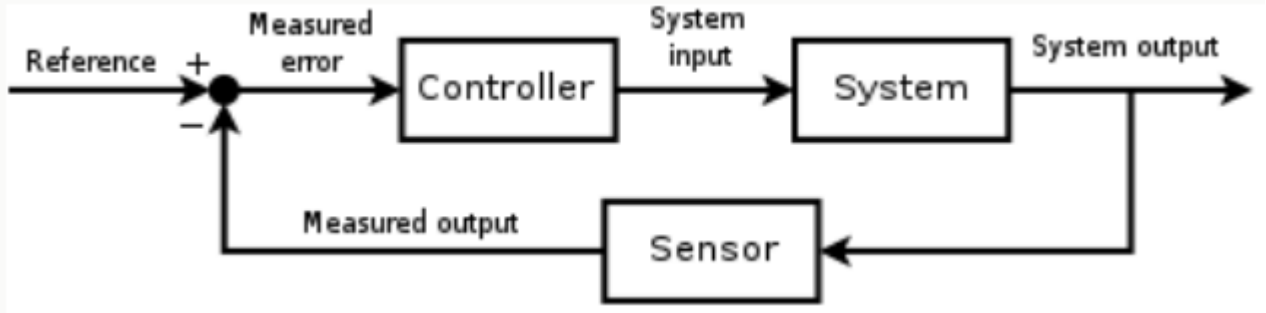
Inertial Sensors - Accelerometers

- A force $\boldsymbol{F}$ is applied to the sensor *box*

- Spring between box and proof mass m
  - Applies force to mass, to compensate applied force
- 

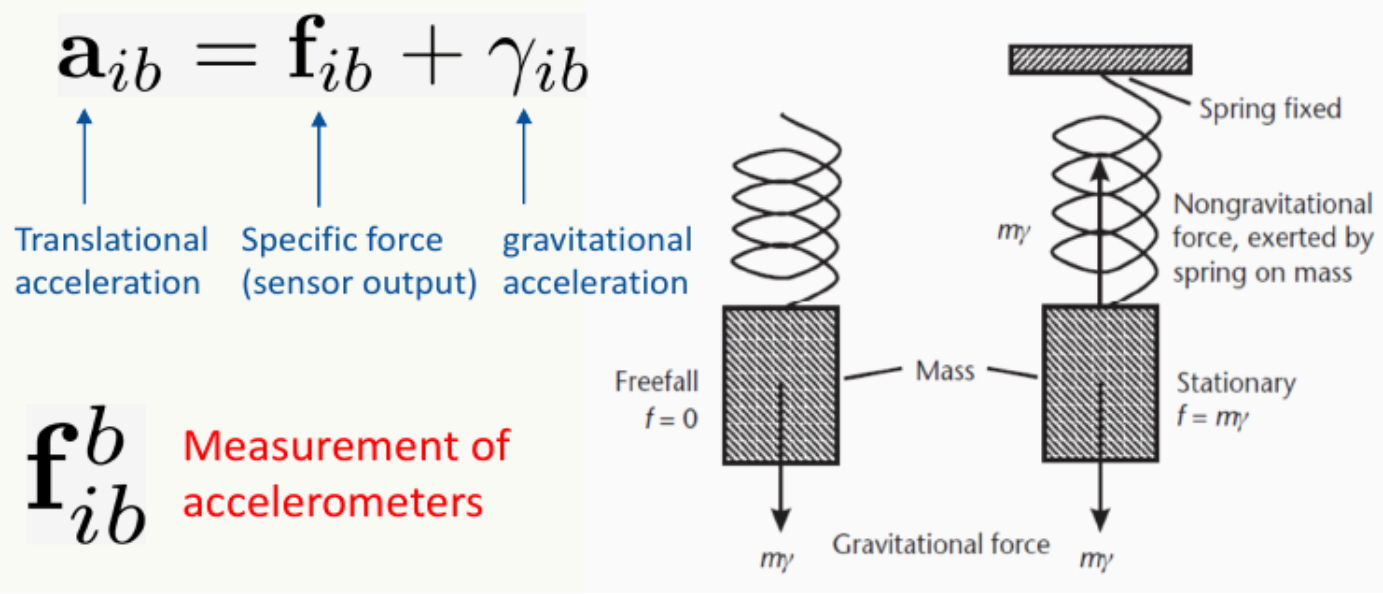
- Hooks law:
  - $F = k \cdot d (= m \cdot a)$
    - k: spring constant
  - $d = \frac{m}{k} \cdot a$
- Displacement measurement via Capacity of Capacitors
  - Applied acceleration ->
  - Distance change ->
  - Capacity change ($C = \frac{A\epsilon_0}{d}$)->
  - Voltage change
- **Closed Loop** Pendulum Accelerometer
  - 

  - Capacitative pickoff observes the displacement
  - The **electromagnet** uses a **compensation current to compensate the displacement**
  - Necessary compensation current is **proportional** to applied acceleration

- 

Measurement of the Accelerometer

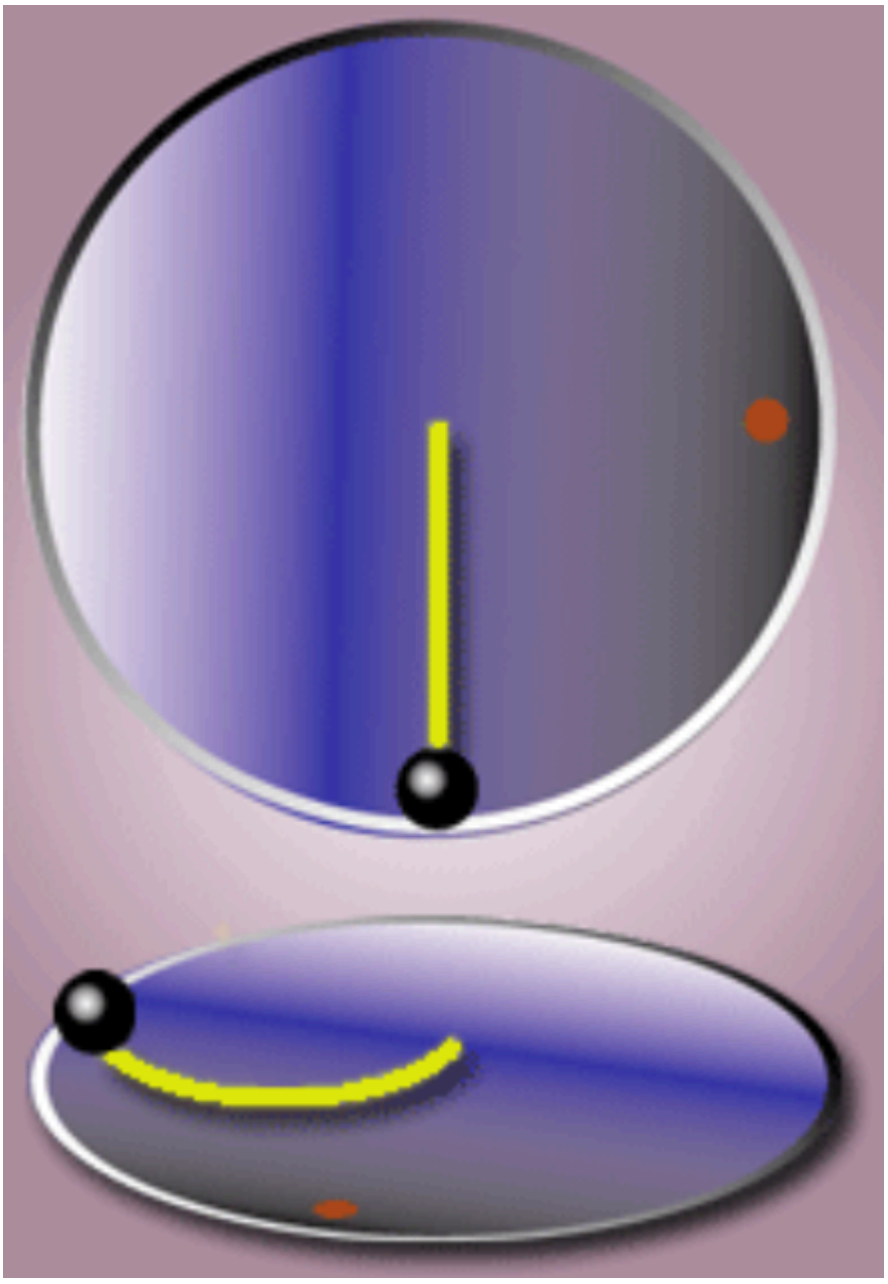
- Non-gravitational force per unit mass (~acceleration) with respect to inertial frame
  - (-> **Specific force** or **proper acceleration**)
    - Or **translational** or **coordinate acceleration**
  - Sensed by sensors and people
  - Acceleration relative to free fall
  - Gravity is property of space
    - -> acts identically on every point of the sensor
      - -> No relative force between box and proof mass, leading to a displacement
    - Displacement detected by a resting sensor is due to something keeping it form free fall (e.g. a table)
- WANTED: Coordinate or translational acceleration (2nd derivative of the position)

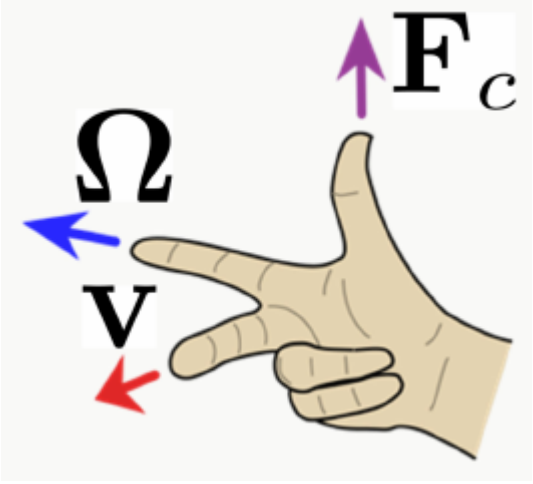


Gyroscope

- The rotation axis of a spinning wheel gyroscope remains constant with respect to the inertial frame
- Rotation applied to the 2nd axis of the spinning wheel induce a rotation on the 3rd axis
- These gyroscope have been used to navigate accross the ocean without GPS and to fly to the moon and back

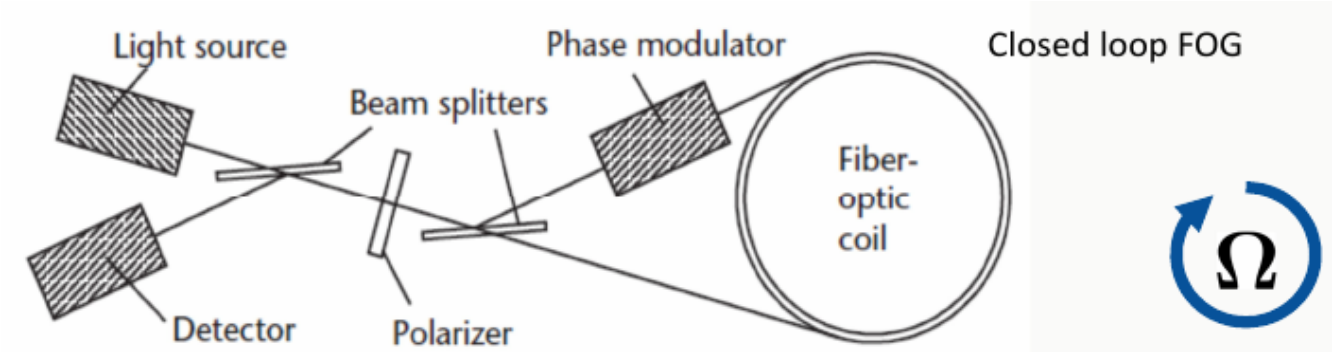
Coriolis Force



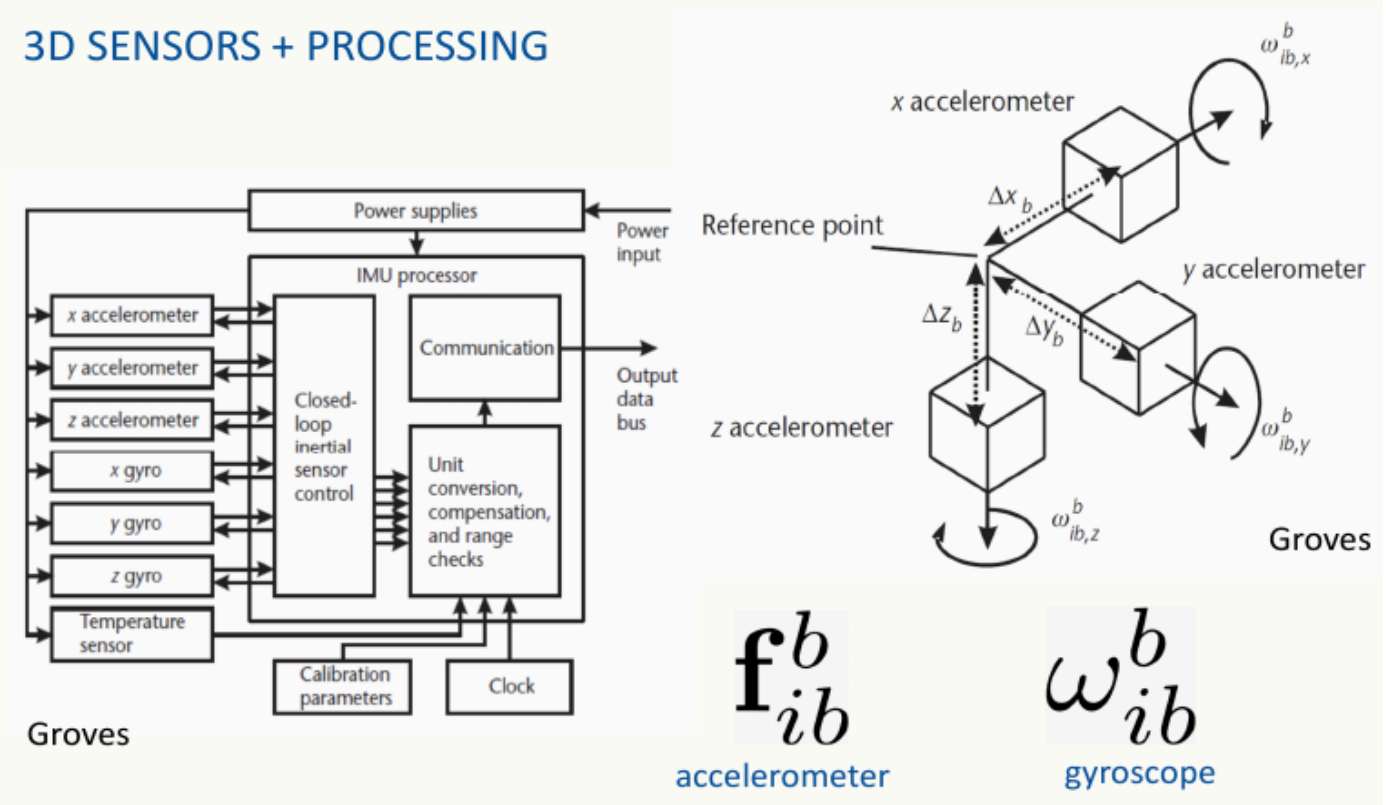


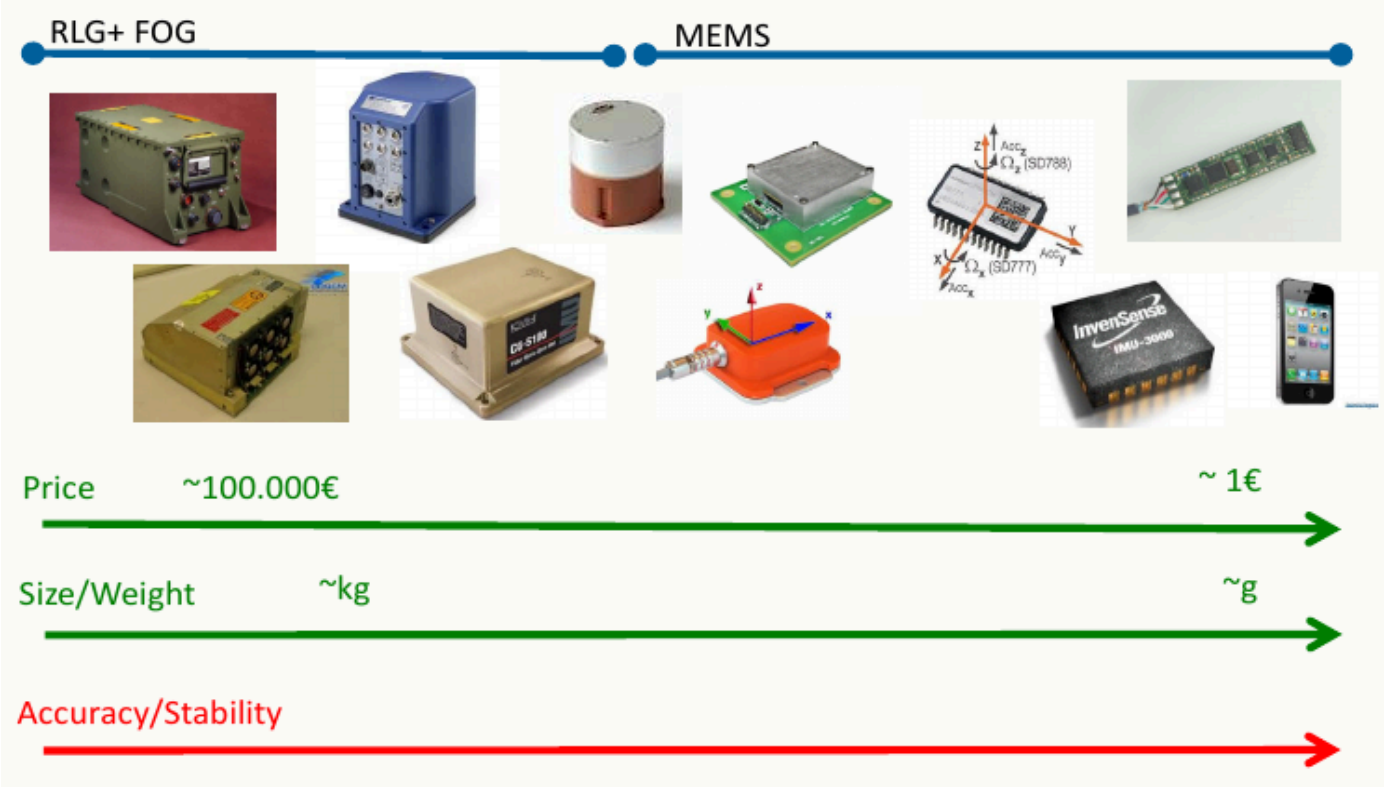
$$F_c = -2m(\Omega \times v)$$

Fiber-Optic Gyro (FOG)

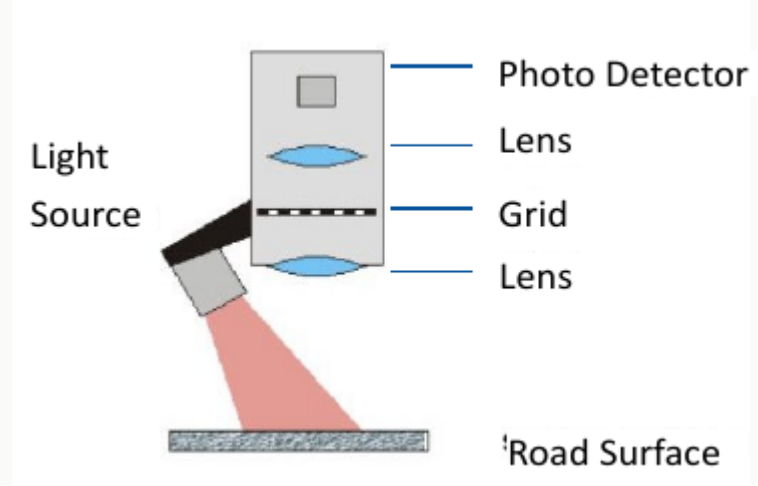
- Light source is splitted and sent via both directions through a fiber-optic coil
- **Sagnac-effect** lead to phase change between the two beams
  - Ring/Coil has different RTT for different direction, if rotating
  - proportional to Ω
  - Bigger area -> higher sensitivity
  - Coil of N rings multiplies the area by N
- Detector measures interference due to phase change
- **Closed loop**: Interference signal is kept constant by introducing an additional phase change in the modular
- 

Inertial Measurement Units (IMU)





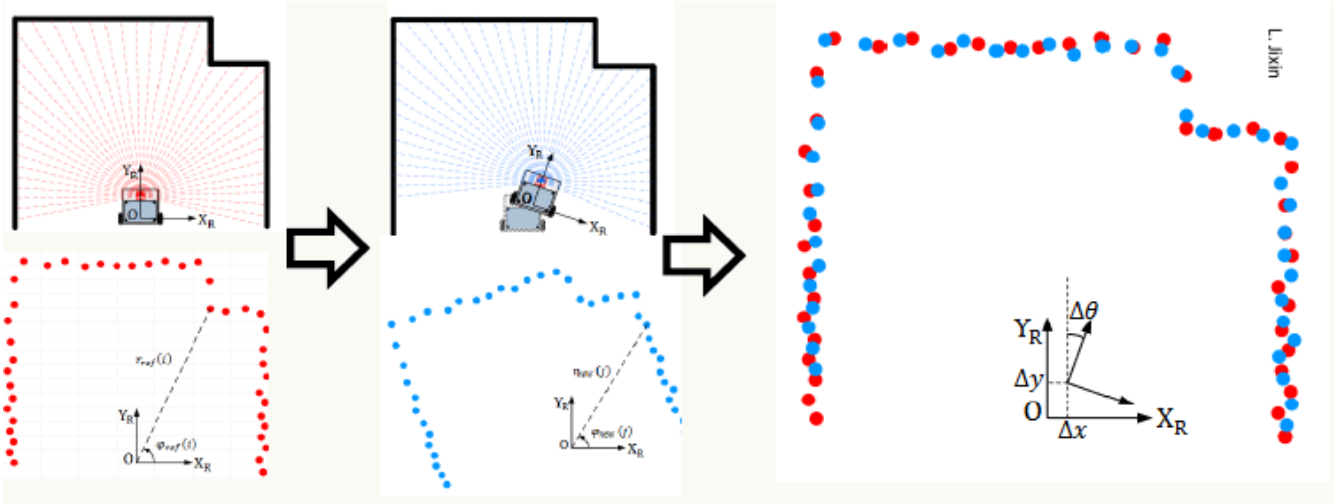
Odometer

- Odometer: Measurement of the travelled distance
- Differential Odometer: Measurement travelled distance at 2 wheels
  - -> additional information about **direction**
- Measurement
  - Photo-electric transducer
  - Magnetband -> magnetic transducer
  - Cog wheel -> magnetic tranducers
- Correlation Based Velocity Measurement
  - 

  - Cross correlation of 2 signals
    - time delay between the signals

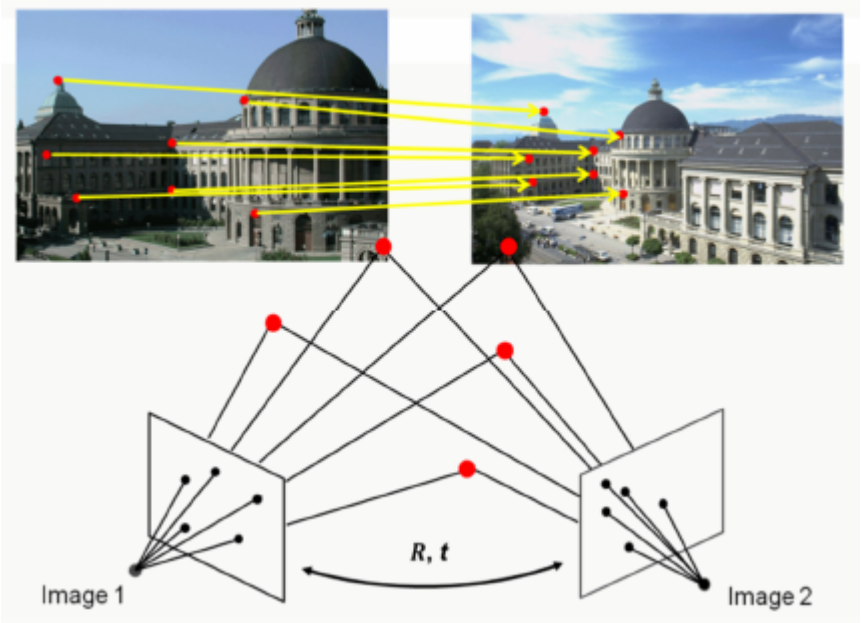
    - velocity of the road surface relative to the sensors

Scan Matching

- Laser Scanned Matching (-> Robotics)
- Assumptions:
    - Static Environment
    - Overlap between the scans
- Usual method:
    - Iterated Closest Point (ICP)
- 

Visual Odometry

- Odometry Based on Images
    - relative motion between two cameras is calculated based on consecutive images
    - Features are calculated in one image
    - Calculate the features in the next images .> find correspondant features in previous image
    - Calculate transformation of the camera between the two images
    - -> relative information
    - -> Drift

- 
Image 1
R, t
Image 2